Jedha

CDSD block 1

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# Project KAYAK

Github project : cyrilbraguy/Bl1_kayak

Kayak application on Huggingface : Kayak_trip_application

# Goal of the project

## Pitch

Design a travel application,
a search engine
**to recommend** the best trip
destinations within a list of
35 cities to visit in France,
from reliable informations on :

Best weather forecast,

hotel rooms availability in
these destinations

## Goal

- Get info on cities : GPS coordinates
- Get weather data of cities at 5 days
  - order cities bv weather results
  - select best 5 cities
- Get hotels info about each destination
- Store all the information above in a data lake : one file .csv
- Extract, transform and load cleaned data from your datalake to a data warehouse
- Display these informations on a nice frontend

# Detailed pipeline :

## 1. Cities + weather

- 35 cities list → get lat & lon from nominatim.org → dataFrame
- Get weather data of cities at 5 days in 3 hrs slots from openweathermap.org [ D to D+4 ]
- Average weather data on each time window eg : [ D+1 – D+4] & [ D+3 – D+4 ]
  - groupby data on each day (average, sum, max)
  - Compute a **weather comfort score** for each trip period
  - order cities by **weather comfort score**
  - select best 5 cities : weather_avg['selected'] =1
- Store in a .csv file on S3
- Comfort score : 30 % * ciel_clair + 25 %*(1-humidite)
  + 20 % (1-haut_pluie) + 15 % (1-(temp_moy-21)/15)
  + 10 % (1-vent_max/15)

# Detailed pipeline :
## 2. Hotels

- booking_scraper.py : A special tool developed to scrape booking.com : using playwright
  - [city, checkin, checkout, nb_adults, nb_children, nb_rooms]
    → 5 hotels w/ several informations :
  - [ hotel_name, url_hotel, address_hotel, hotel_description, lat_hotel, lon_hotel, score_hotel, price_hotel, currency_hotel, room_description, hotel_city, combined_score_hotel ]
- Combined_score = score_hotel - p * (price_hotel - max_price).clip(lower=0)
- Store all the information above in a data lake : **hotels_top20_cities.csv**

# Detailed pipeline

## 3. ETL & SQL database

- Load last **hotels_top20_cities.csv**
- Save information in 3 sql tables to respect unicity principle :
  cities  weather  hotels

| table | key_id | columns |
|---|---|---|
| cities | city_id | city, city_id, city_lat, city_lon, city_name_, city_plus |
| weather | city_id ; checkin_date ; checkout_date | city_id, city, checkin_date, checkout_date, avg_comfort_score, ... selected, lat, lon |
| hotels | city_id ; checkin_date ; checkout_date | city_id, city, checkin_date, ... combined_score_hotel |

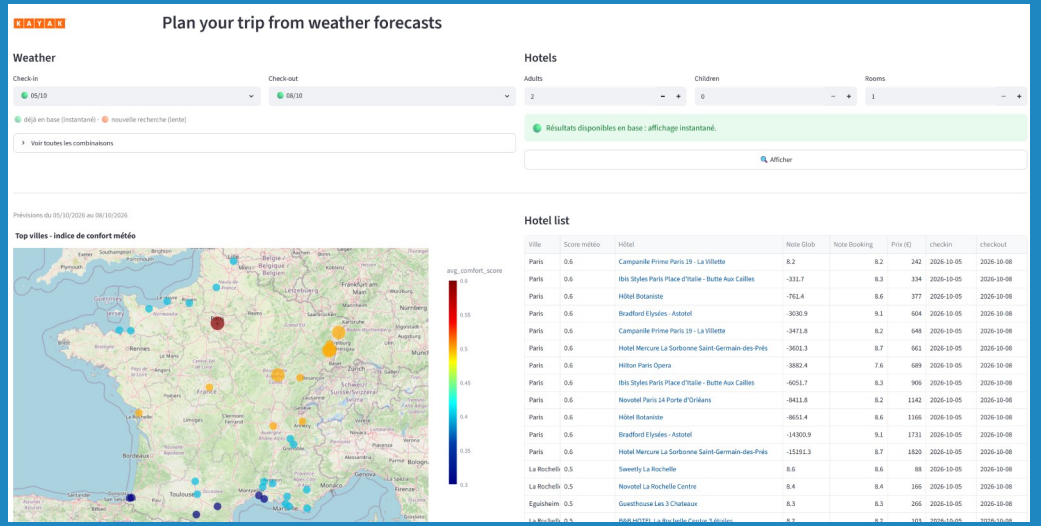
- A sql database on Neondb is used

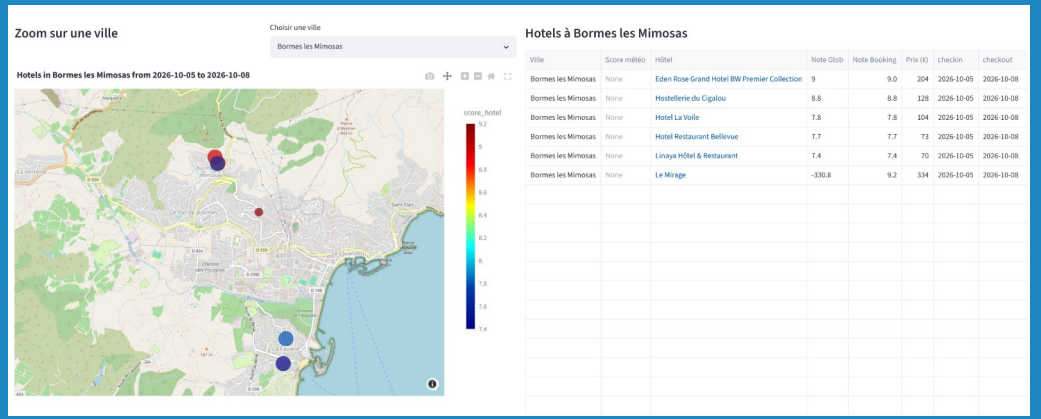
# Detailed pipeline

## 4. Frontend

A streamlit frontend is used

- Reads sql database and displays filtered and ordered data
- 1 : weather & best cities
  displays best cities w/ color map & bigger marker size
  & hotels listed by order of note_glob
- 2 : select one cities ; view hotels list order by combined_score



# Complete solution

**An application in a Docker container
in huggingface space**

- 1 : pipeline linear get weather hotels ;
  ETL : cron 6h & 18 h ; store in S3 & neon SQL dB
- 2 : Streamlit frontend to print results for customer
- Git repo for code store & version
- Github_action to deploy in HuggingFace space at push
- S3 bucket Sql database on NeondB

demo

# Questions ?